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Assignment No 3

Employee Performance Appraisal System

1) Single - Row Subquery:

```
mysql> select Emp_Name,Salary from employee where salary=(select max(salary)
from employee);
+----------+--------+
| Emp_Name | Salary |
+----------+--------+
| Mahadev  |  90000 |
+----------+--------+
1 row in set (0.00 sec)
```

```
mysql> select Emp_Name from employee where salary=(select max(salary)from em
ployee);
+----------+
| Emp_Name |
+----------+
| Mahadev  |
+----------+
1 row in set (0.00 sec)
```

```
mysql> select Emp_Name from employee where salary=(select min(salary)from em
ployee);
+----------+
| Emp_Name |
+----------+
| Sahil    |
+----------+
1 row in set (0.00 sec)
```

```
mysql> select Emp_Name from employee where salary!=(select min(Salary)
from employee);
+----------+
| Emp_Name |
+----------+
| Mahadev  |
| Abhishek |
| Mahesh   |
+----------+
3 rows in set (0.00 sec)
```

2). Multiple - Row Subquery:

```
mysql> SELECT Emp_Name
    -> FROM employee
    -> WHERE Dept_Id IN (SELECT Dept_Id FROM employee WHERE Salary > 30000);
+----------+
| Emp_Name |
+----------+
| Rahul    |
| Vikas    |
| Priya    |
| Sneha    |
| Arjun    |
+----------+
5 rows in set (0.01 sec)
```

3).Exists Operator:

```
mysql> SELECT Emp_Name
    -> FROM employee e
    -> WHERE EXISTS (SELECT 1 FROM employee WHERE Dept_Id = e.Dept_Id AND Salary > 30000);
+----------+
| Emp_Name |
+----------+
| Rahul    |
| Vikas    |
| Priya    |
| Sneha    |
| Arjun    |
+----------+
5 rows in set (0.00 sec)
```

4) Multiple Column Subquery:

```
mysql> SELECT Emp_Name, Salary
    -> FROM employee
    -> WHERE (Dept_Id, Salary) IN (SELECT Dept_Id, MAX(Salary) FROM employee GROUP BY Dept_Id);
+----------+--------+
| Emp_Name | Salary |
+----------+--------+
| Vikas    |  42000 |
| Priya    |  36000 |
| Sneha    |  52000 |
+----------+--------+
3 rows in set (0.01 sec)
```

5) Correlated Subquery:

```
mysql> SELECT Emp_Name, Salary
    -> FROM employee e1
    -> WHERE Salary > (SELECT AVG(Salary) FROM employee e2 WHERE e1.Dept_Id = e2.Dept_Id);
+----------+--------+
| Emp_Name | Salary |
+----------+--------+
| Vikas    |  42000 |
| Priya    |  36000 |
+----------+--------+
2 rows in set (0.00 sec)
```

6) Nested Subquery:

```
mysql> SELECT Emp_Name
    -> FROM employee
    -> WHERE Salary = (SELECT MAX(Salary) FROM (SELECT Salary FROM employee) AS temp);
+----------+
| Emp_Name |
+----------+
| Sneha    |
+----------+
1 row in set (0.00 sec)
```

7) Joins:

i) Inner Join:

```
mysql> SELECT e.Emp_Name, d.Dept_Name
    -> FROM employee e
    -> INNER JOIN department d
    -> ON e.Dept_Id = d.Dept_Id;
+----------+-----------+
| Emp_Name | Dept_Name |
+----------+-----------+
| Rahul    | Sales     |
| Priya    | Sales     |
| Vikas    | Marketing |
| Arjun    | Marketing |
| Sneha    | Accounts  |
+----------+-----------+
5 rows in set (0.00 sec)
```

ii) Left Join:

```
mysql> SELECT e.Emp_Name, d.Dept_Name
    -> FROM employee e
    -> LEFT JOIN department d
    -> ON e.Dept_Id = d.Dept_Id;
+----------+-----------+
| Emp_Name | Dept_Name |
+----------+-----------+
| Rahul    | Sales     |
| Vikas    | Marketing |
| Priya    | Sales     |
| Sneha    | Accounts  |
| Arjun    | Marketing |
+----------+-----------+
5 rows in set (0.00 sec)
```

iii) Right Join:

```
mysql> SELECT e.Emp_Name, d.Dept_Name
    -> FROM employee e
    -> RIGHT JOIN department d
    -> ON e.Dept_Id = d.Dept_Id;
+----------+-----------+
| Emp_Name | Dept_Name |
+----------+-----------+
| Rahul    | Sales     |
| Priya    | Sales     |
| Vikas    | Marketing |
| Arjun    | Marketing |
| Sneha    | Accounts  |
+----------+-----------+
5 rows in set (0.00 sec)
```

iv) Full Join

```
mysql> SELECT e.Emp_Name, d.Dept_Name
    -> FROM employee e
    -> LEFT JOIN department d
    -> ON e.Dept_Id = d.Dept_Id
    ->
    -> UNION
    ->
    -> SELECT e.Emp_Name, d.Dept_Name
    -> FROM employee e
    -> RIGHT JOIN department d
    -> ON e.Dept_Id = d.Dept_Id;
+----------+-----------+
| Emp_Name | Dept_Name |
+----------+-----------+
| Rahul    | Sales     |
| Vikas    | Marketing |
| Priya    | Sales     |
| Sneha    | Accounts  |
| Arjun    | Marketing |
+----------+-----------+
5 rows in set (0.01 sec)
```

v) Cross Join:

```
mysql> SELECT e.Emp_Name, d.Dept_Name
    -> FROM employee e
    -> CROSS JOIN department d;
+----------+-----------+
| Emp_Name | Dept_Name |
+----------+-----------+
| Rahul    | Accounts  |
| Rahul    | Marketing |
| Rahul    | Sales     |
| Vikas    | Accounts  |
| Vikas    | Marketing |
| Vikas    | Sales     |
| Priya    | Accounts  |
| Priya    | Marketing |
| Priya    | Sales     |
| Sneha    | Accounts  |
| Sneha    | Marketing |
| Sneha    | Sales     |
| Arjun    | Accounts  |
| Arjun    | Marketing |
| Arjun    | Sales     |
+----------+-----------+
15 rows in set (0.00 sec)
```

vi) Self Join:

```
mysql> SELECT e1.Emp_Name, e2.Emp_Name
    -> FROM employee e1, employee e2
    -> WHERE e1.Dept_Id = e2.Dept_Id AND e1.Emp_id <> e2.Emp_id;
+----------+----------+
| Emp_Name | Emp_Name |
+----------+----------+
| Priya    | Rahul    |
| Arjun    | Vikas    |
| Rahul    | Priya    |
| Vikas    | Arjun    |
+----------+----------+
4 rows in set (0.00 sec)
```

8) Comparison Operators:

```
mysql> SELECT * FROM employee WHERE Salary > 20000;
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    101 | Rahul    |  32000 |       1 |
|    102 | Vikas    |  42000 |       2 |
|    103 | Priya    |  36000 |       1 |
|    104 | Sneha    |  52000 |       3 |
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM employee WHERE Salary < 50000;
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    101 | Rahul    |  32000 |       1 |
|    102 | Vikas    |  42000 |       2 |
|    103 | Priya    |  36000 |       1 |
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM employee WHERE Salary = 30000;
Empty set (0.00 sec)
```

9) Logical Operators:

```
mysql> SELECT * FROM employee WHERE Salary > 20000 AND Dept_Id = 1;
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    101 | Rahul    |  32000 |       1 |
|    103 | Priya    |  36000 |       1 |
+--------+----------+--------+---------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM employee WHERE Salary > 20000 OR Dept_Id = 2;
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    101 | Rahul    |  32000 |       1 |
|    102 | Vikas    |  42000 |       2 |
|    103 | Priya    |  36000 |       1 |
|    104 | Sneha    |  52000 |       3 |
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM employee WHERE NOT Dept_Id = 1;
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    102 | Vikas    |  42000 |       2 |
|    105 | Arjun    |  27000 |       2 |
|    104 | Sneha    |  52000 |       3 |
+--------+----------+--------+---------+
3 rows in set (0.00 sec)
```

10) Range Operators:

```
mysql> SELECT * FROM employee WHERE Salary BETWEEN 20000 AND 50000;
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    101 | Rahul    |  32000 |       1 |
|    102 | Vikas    |  42000 |       2 |
|    103 | Priya    |  36000 |       1 |
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
4 rows in set (0.00 sec)
```

11) Set Operators:

```
mysql> SELECT Emp_Name FROM employee
    -> UNION
    -> SELECT Emp_Name FROM manager;
+----------+
| Emp_Name |
+----------+
| Rahul    |
| Vikas    |
| Priya    |
| Sneha    |
| Arjun    |
+----------+
5 rows in set (0.00 sec)
```

12) Pattern Matching Operators:

```
mysql> SELECT * FROM employee WHERE Emp_Name LIKE 'A%';
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
1 row in set (0.01 sec)

mysql> SELECT * FROM employee WHERE Emp_Name LIKE '%n';
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
1 row in set (0.00 sec)
```

13) Null Operators:

```
mysql> SELECT * FROM employee WHERE Salary IS NULL;
Empty set (0.00 sec)

mysql> SELECT * FROM employee WHERE Salary IS NOT NULL;
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    101 | Rahul    |  32000 |       1 |
|    102 | Vikas    |  42000 |       2 |
|    103 | Priya    |  36000 |       1 |
|    104 | Sneha    |  52000 |       3 |
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
5 rows in set (0.00 sec)
```

14) Subquery Operators:

```
mysql> SELECT * FROM employee WHERE Salary > ALL (SELECT Salary FROM employe
e WHERE Dept_Id = 1);
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    102 | Vikas    |  42000 |       2 |
|    104 | Sneha    |  52000 |       3 |
+--------+----------+--------+---------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM employee WHERE Salary > ANY (SELECT Salary FROM employe
e WHERE Dept_Id = 1);
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    102 | Vikas    |  42000 |       2 |
|    103 | Priya    |  36000 |       1 |
|    104 | Sneha    |  52000 |       3 |
+--------+----------+--------+---------+
3 rows in set (0.00 sec)
```

15) Boolean Operators:

```
mysql> SELECT * FROM employee WHERE (Salary > 20000) = TRUE;
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    101 | Rahul    |  32000 |       1 |
|    102 | Vikas    |  42000 |       2 |
|    103 | Priya    |  36000 |       1 |
|    104 | Sneha    |  52000 |       3 |
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
5 rows in set (0.00 sec)
```

16) Regular Expressions:

```
mysql> SELECT * FROM employee WHERE Emp_Name REGEXP '^A';
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
1 row in set (0.04 sec)

mysql> SELECT * FROM employee WHERE Emp_Name REGEXP 'n$';
+--------+----------+--------+---------+
| Emp_id | Emp_Name | Salary | Dept_Id |
+--------+----------+--------+---------+
|    105 | Arjun    |  27000 |       2 |
+--------+----------+--------+---------+
1 row in set (0.00 sec)
```

17) Arithmetic Operators:

```
mysql> SELECT Salary, Salary + 1000 AS Bonus FROM employee;
+--------+-------+
| Salary | Bonus |
+--------+-------+
|  32000 | 33000 |
|  42000 | 43000 |
|  36000 | 37000 |
|  52000 | 53000 |
|  27000 | 28000 |
+--------+-------+
5 rows in set (0.01 sec)

mysql> SELECT Salary, Salary * 2 FROM employee;
+--------+------------+
| Salary | Salary * 2 |
+--------+------------+
|  32000 |      64000 |
|  42000 |      84000 |
|  36000 |      72000 |
|  52000 |     104000 |
|  27000 |      54000 |
+--------+------------+
5 rows in set (0.00 sec)

mysql> SELECT Salary, Salary - 500 FROM employee;
+--------+--------------+
| Salary | Salary - 500 |
+--------+--------------+
|  32000 |        31500 |
|  42000 |        41500 |
|  36000 |        35500 |
|  52000 |        51500 |
|  27000 |        26500 |
+--------+--------------+
5 rows in set (0.00 sec)
```